Year:2025-26
Semester: II

1. **Name of the Faculty: Amit Verma** **Course Code: CSEG1043**
2. **Course** : **Data Structures and Algorithms** **L: 4**
3. **Program** : B.Tech (CSE-All Branches) ,B.C.A and B.Sc **T: 0**
4. **Target** : 60% **P: 0**

C: 4

COURSE PLAN

Target	50% (Marks)
Level-1	40% (Population)
Level-2	40% (Population)
Level-3	50% (Population)

1. Method of Evaluation

Evaluation Component	Sub-Component	Marks / Weightage (%)
	THEORY ASSESSMENT	
End Semester Examination	**End Sem Exam**	**30%**
Mid Semester Examination	**Mid Sem Exam**	**20%**
Internal Assessment (IA)	**Total IA**	**50%**
	Theory IA	**25%**
	Quiz + Test – 1	**7.5**
	Quiz + Test – 2	**7.5**
	Assignment – 1	**5**
	Assignment – 2	**5**
	LAB ASSESSMENT	
	Lab IA	**25%**
	Continuous Lab Assessment	**10**
	Viva – 1	**5**
	Viva – 2	**5**
	Hundred Days of Data Structures	**5**
Total Evaluation		**100%**

2. Passing Criteria

UPES

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Scale	**UG**
Out of 10 Point Scale	SGPA – "5.0" in Each Semester CGPA – "5.0" Min. Individual Course Grade – "C" Course Grade Point – "4.0"

*for UG, passing marks are 35/100 in a paper

3. **Pedagogy**
   - Face to Face (Context-Based Learning)
   - Online lectures(Adaptive Teaching)
   - Class Test
   - Video Lectures
   - Presentation
   - Concept Dairy (Student maintain the record of content which they understood from the sessions)

4. **References:**

Text Books	Web resources	Journals	Reference books
1. S. Lipschutz, "Data Structures with C", Schaum's Outline Series, McGraw-Hill Education (India) Pvt. Limited, 2017. 2. Y. P. Kanetkar, "Data structures through C", 4rd Edition, New Delhi: BPB, 2022.	https://ocw.mit.edu/courses/civil-and-environmental-engineering/1-204-computer-algorithms-in-systems-engineering-spring-2010/lecture-notes/MIT1_204S10_lec05.pdf https://cse.iitkgp.ac.in/~pb/algo1-pb-101031.pdf https://www.cs.cmu.edu/~rjsimmon/15122-s16/lec/10-linkedlist.pdf https://cse.iitrpr.ac.in/ckn/courses/f2015/csl201/w1.pdf		1. A. V. Aho, J. E. Hopcroft, and J. D. Ullman, "Data Structures and Algorithms", New Delhi: Pearson Education, 2003. 2. E. Horowitz, and S. Sahni, "Fundamentals of Data Structures in C", 2nd Edition, Hyderabad: University Press, 2008.

GUIDELINES TO STUDY THE SUBJECT

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Instructions to Students:

1. Go through the 'Syllabus' in the LMS section of the web-site (https://myupes-beta.upes.ac.in) in order to find out the Reading List.
2. Get your schedule and try to pace your studies as close to the timeline as possible.
3. Get your on-line lecture notes (Content, videos) at Lecture Notes section. These are our lecture notes. Make sure you use them during this course.
4. Check your LMS regularly
5. Go through study material
6. Check mails and announcements on LMS
7. Keep updated with the posts, assignments and examinations which shall be conducted on the LMS
8. Be regular, so that you do not suffer in any way
9. **Cell Phones and other Electronic Communication Devices:** Cell phones and other electronic communication devices (such as Blackberries/Laptops) are not permitted in classes during Tests. Such devices MUST be turned off in the class room.
10. **e-Mail and online learning tool:** Each student in the class should have an e-mail id and a pass word to access the LMS system regularly. Regularly, important information – Date of conducting class tests, guest lectures, via online learning tool. The best way to arrange meetings with us or ask specific questions is by email and prior appointment. All the assignments preferably should be uploaded on online learning tool. Various research papers/reference material will be mailed/uploaded on online learning platform time to time.
11. **Attendance:** Students are required to have minimum attendance of 80% in each subject.

This much should be enough to get you organized and on your way to having a great semester! If you need us for anything, send your feedback through e-mail to your concerned faculty. Please use an appropriate subject line to indicate your message details.

There will no doubt be many more activities in the coming weeks. So, to keep up to date with all the latest developments, please keep visiting this website regularly.

RELATED OUTCOMES

1. The expected outcomes of the Program are:

PO1	*Engineering Knowledge*: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	*Problem Analysis*: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	*Design/Development of Solutions*: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	*Conduct Investigations of Complex Problems*: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis

	of the information to provide valid conclusions.
PO5	*Modern Tool Usage*: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	*The Engineer and Society*: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	*Environment and Sustainability*: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	*Ethics*: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	*Individual and Team-work*: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	*Communication*: Communicate effectively on complex engineering activities with the engineering community and with society at-large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	*Project Management and Finance*: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	*Life-long Learning*: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

2. **The expected outcomes of the Specific Program are: (upto3)**

3. **The expected outcomes of the Course are: (minimum 3 and maximum 6)**

CO 1	State the significance and properties of the fundamental data structures.
CO 2	Implement common data structures while ensuring proper memory management and error handling.
CO 3	Illustrate expertise in understanding the common sorting and searching techniques with their complexities and implement them.
CO 4	Analyse real-world problems by understanding the trade-offs involved in identifying the appropriate data structure(s) based on problem requirements and using them to solve the problems efficiently.

4. **Co-Relationship Matrix**

Indicate the relationships by1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

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Program Outcomes / Course Outcomes	PO 1	PO 2	PO 3	PO4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO 1	1		2	1								1			
CO 2	1	1	2	1											
CO 3	1	1	2	1								2			
CO 4	1	2	2	2								2			
Average	1	1	2	1.25								1.25			

5. **Course outcomes assessment plan:**

components / Course Out-	CP-1	CP-2	CP-3	Any other
CO 1	√	√	√	£
CO 2	√	√	√	£
CO 3	√	√	√	£
CO 4	√	√	√	£

BROAD PLAN OF COURSE COVERAGE

Course Activities:

S.	Description	Planned	Re-

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No.		From	To	No. of Sessions	marks
1.	Unit I: INTRODUCTION			12 Lecture Hours	
2.	Unit II: LINKED LIST			10 Lecture Hours	
3.	Unit III: STACK & QUEUE			10 Lecture Hours	
4.	Unit IV: TREE			10 Lecture Hours	
5.	Unit V: HASH TABLE & GRAPH			10 Lecture Hours	
6.	Unit VI: SORTING & SEARCHING			8 Lecture Hours	
Total Session				60	

SESSION PLAN

UNIT – I (12 Lectures)

Lecture No.	Topics to be Covered	CO Mapped
1	Overview, classification, importance of data structures	1
2	Role of data structures in programming/problem solving	1
3	Basic terminology: elements, operations, storage	1
4	Memory allocation, garbage collection, compaction	1
5	Iterative approach	1
6	Recursive approach	1
7	Asymptotic analysis	1

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Lecture No.	Topics to be Covered	CO Mapped
8	Amortized analysis	1
9	Array: 1D & 2D memory representation	1
10	Array operations: insertion, deletion	1
11	Array operations: searching and applications	1
12	Structures: nested, function pointer, self-referential, anonymous unions, ADTs	1

UNIT – II (10 Lectures)

Lecture No.	Topics to be Covered	CO Mapped
13	Singly Linked List and operations	2
14	Doubly Linked List and operations	2
15	Circular Linked List and operations	2
16	Header List and Sentinel Node	2
17	Generalized Linked List	2
18	Skip List	2
19	Applications of Linked Lists	2
20	Polynomial manipulation using Linked List	2
21	Implementation of other data structures using Linked List	2
22	Revision/Problem discussion on Linked Lists	2

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UNIT – III (10 Lectures)

Lecture No.	Topics to be Covered	CO Mapped
23	Stack data structure and operations	2
24	Queue data structure and operations	2
25	Implementation of Stack & Queue using Array	2
26	Implementation of Stack & Queue using Linked List	2
27	Circular Queue	2
28	Deque and its types	2
29	Priority Queue	2
30	Stack applications: Infix to Prefix/Postfix	2
31	Expression evaluation, note on DFS	2
32	Queue applications: Job scheduling, note on BFS	2

UNIT – IV (10 Lectures)

Lecture No.	Topics to be Covered	CO Mapped
33	Introduction to Trees and terminology	3
34	Binary Tree properties	3
35	Tree traversals (level, in, pre, post)	3
36	Threaded Binary Tree	3
37	BST: properties and operations	3

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Lecture No.	Topics to be Covered	CO Mapped
38	Balanced BST concept	3
39	AVL Tree: properties & rotations	3
40	AVL Tree: insertion & deletion	3
41	Red-Black Tree	3
42	Multi-way Search Tree & B-Tree basics	3

UNIT – V (10 Lectures)

Lecture No.	Topics to be Covered	CO Mapped
43	Hashing and hash functions	3
44	Hash table structure & collisions	3
45	Collision resolution techniques	3
46	Maintaining load factor	3
47	Applications of hash tables	3
48	Introduction to Graph and terminology	3
49	Graph representations: adjacency matrix & list	3
50	Graph traversal: DFS and BFS	3
51	Connected Components	3
52	Minimum Spanning Tree & Shortest Path	3

UPES

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C: 4

UNIT – VI (8 Lectures)

Lecture No.	Topics to be Covered	CO Mapped
53	Stability and in-place properties	4
54	Internal vs external sorting	4
55	Bubble, selection, insertion sort	4
56	Lower bound for comparison-based sorting	4
57	Recursive merge sort	4
58	Recursive quicksort	4
59	Recursive binary search	4
60	Complexities of sorting and searching algorithms	4

Data Structures and Algorithms Lab

List of Experiments

Experiment 1: Basic Data Structure

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			C: 4

To demonstrate the use of array, structure, and union along with dynamic memory allocation.

Experiment 2: Link List Data Structure and its Applications

To experiment with the concept of pointers, structure, and dynamic memory allocation to realize linked lists, their types, and application.

Experiment 3: Stack Data Structure

To use arrays and linked lists to implement Stack and its applications.

Experiment 4: Queue Data Structure

To demonstrate the use of arrays and linked lists to implement different variants of Queue and its applications.

Experiment 5: Trees

To demonstrate the creation of a binary tree and working with tree traversal.

Experiment 6: Heaps

To create a heap data structure and implement its operations, and its applications.

Experiment 7: Hash Tables

To implement a hash table using various collision resolution techniques, and its applications.

Experiment 8: Graphs

To demonstrate the creation of graphs and working with graph traversal algorithms.

Experiment 9: Sorting algorithms

To implement common sorting algorithms.

Experiment 10: Searching algorithms

To implement various search algorithms on data structures.